Supplementary Material for Diagnosing and Repairing Out-of-Distribution Failure in MOBA Draft Policies

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TABLE I
MCTS TRAINING THROUGHPUT ACROSS IMPLEMENTATION STAGES (THE SUBMISSION'S SEARCH).

Implementation	ep/s	Bottleneck
Python + PyTorch CPU	1.2	Forward pass dispatch
C++ tree, batched virtual loss (K=32)	5.0	Amortized sync
Full CUDA kernel (1 block=1 ep)	26	Python buffer overhead
+ ring buffer + pipelining (bring-up)	50	Warm-up, small batches
Final (steady state, campaign)	**140**	GPU compute

I. FUSED CUDA KERNELS FOR MCTS TRAINING

The MCTS training campaign in the main paper, 227 complete training runs (the submission's 195 and the revision's 32) spanning simulation budgets up to 800 per move and episode counts up to 1M, was only feasible because of the fused CUDA implementation described here. AlphaZero-style MCTS training is computationally demanding: each episode requires $\sim$8 MCTS searches of 200 simulations each, with a neural network forward pass at every leaf node. Standard implementations incur Python interpreter overhead, CUDA kernel launch latency, and host-device synchronization for each of the $\sim$3,200 forward passes per episode.

We developed a fused CUDA implementation that runs entire MCTS episodes inside a single kernel launch. One thread block (256 threads) executes one complete draft episode: thread 0 drives all sequential logic (UCB selection, action sampling, backpropagation) while all 256 threads cooperate on matrix multiplications during neural network forward passes via shared memory. The forward pass functions (`linear_layer`, `batchnorm_relu`, `residual_block`) are compiled as `__device__` functions callable from within the kernel with zero launch overhead. Two additional optimizations compound with the fused kernel:

- **Split backbone from heads**: Value symmetrization runs only the small value head (257→128→64→1) twice rather than the full backbone, halving symmetrization cost.
- **Pipelined generation**: Episode results are written into pre-allocated ring buffers via C++ `memcpy`; a generation thread launches GPU batches asynchronously while the main thread performs gradient updates.

Table I shows the progression; all stages measured at 200 simulations per move. The first four rows are development-time measurements taken as each optimization landed; the final row is the sustained steady-state rate of the finished implementation, measured across the submission's training campaign (195 complete runs, 128 concurrent episodes per kernel launch) under four-GPU load on NVIDIA RTX PRO 6000 Blackwell Workstation Edition GPUs (96 GB each). The highest rate recorded in the training logs is 161 episodes/second. The rows are heterogeneous in leaf evaluation (development history, not a controlled comparison); the controlled comparison appears below. The gap between the bring-up and final figures reflects cumulative-average warm-up in early measurements and subsequent batching maturity. Sustained throughput is 117$\times$ the Python baseline and scales inversely with simulation budget (74 episodes/second at 400 simulations per move), so the implementation is limited by compute, with little fixed overhead.

a) The search used in this revision.: The submission's search expanded a node only at our team's turns, so its tree covered the current own-pick block. Every agent in this revision trains with a search that also expands the opponent's turns: an opponent node is a chance node whose GD reply distribution is computed once and cached, its children are added with progressive widening (at most $\lceil\sqrt{N}\rceil$ after N visits), and each simulation creates at most one new own decision, which costs a policy forward pass. Trees are allocated per episode ($3\times$ simulations $+8$ nodes). We re-measured the kernel with the protocol of the controlled comparison below (128 concurrent episodes per launch, one warm-up launch, then 768 timed episodes; the paper's operating-point agent and leaf WP) on one NVIDIA RTX 5090, because the original GPUs were not available. With the revision's search the kernel runs 63 episodes/second at 200 simulations per move and 33 at 400; with the submission's search on the same GPU, 216 and 112. The deeper search costs a factor of 3.4 in throughput. A 300K-episode run at 400 simulations takes about 3.7 hours on one of the original GPUs.

b) Apples-to-apples baseline (submission's search).: For a controlled speedup claim we built a tuned host-tree baseline in the style of KataGo's batched-evaluation pipelines: 64–256 CPU threads each running one episode's tree, leaf evaluations batched across threads onto the GPU via cuBLAS GEMM over multiple streams, reproducing the production kernel's exact

TABLE II
STRENGTH-ADJUSTED VALIDITY OF THE INTERACTION METRICS: ACTUAL WIN RATE IN THE TOP MINUS THE BOTTOM METRIC DECILE (PP), AND THE NUMBER OF INCREASING ADJACENT DECILE STEPS (OF 9). DECILES ARE COMPUTED WITHIN STRATA OF SKILL TIER × OWN-TEAM × OPPOSING-TEAM MEAN-HERO-WR QUINTILES AND COUNT-WEIGHTED ACROSS STRATA (95% CIS ABOUT ±1.1PP ON THE TEST SET AND ±0.4PP POST-SNAPSHOT). "SELF-INCL.": THE STATISTICS CONTAIN THE SCORED GAMES' OWN OUTCOMES.

Statistics	Games	Synergy	Counter	Self-incl.
External	test (77,738)	+11.3, 9/9	+11.0, 8/9	yes
External	post (521,078)	+8.8, 9/9	+9.2, 9/9	no
Own train split	test	+9.5, 7/9	+14.9, 7/9	no
Own train split	post	+7.5, 9/9	+12.6, 9/9	no

TABLE III
SYNERGY AND COUNTER RESPONSIVENESS OF THE MCTS CONFIGURATIONS' BENCHMARK DRAFTS UNDER OUR TRAINING-SPLIT STATISTICS (OWN) AND THE EXTERNAL AGGREGATES (EXT.). MEAN OVER SEEDS.

	Synergy		Counter	
Config	Own	Ext.	Own	Ext.
B	+0.93	+1.06	+.074	−.028
F	+1.08	+1.29	+.087	−.020
J	+1.06	+1.31	+.120	+.023
E (1M ep.)	+1.20	+1.21	+.082	−.022
K (no feat.)	+0.88	+1.24	+.116	+.016
B, in-sample	+1.14	+1.07	+.086	−.015
F, in-sample	+1.74	+1.56	+.091	+.006
J, in-sample	+2.38	+2.02	+.094	+.009

per-simulation work (policy priors at expansion, Generic Draft rollout to terminal at every leaf, symmetrized enriched-WP scoring). At equal work *and* equal search semantics (leaf batch 1; first-turn root-visit total-variation distance from the kernel 0.047, at the 0.044 kernel-vs-kernel seed-noise floor; win rate vs. GD 0.761 vs. the kernel's 0.757), the host pipeline sustains 68 ep/s against the fused kernel's 151 ep/s on the same GPU: the fused kernel is 2.2× faster, the figure quoted in the main paper, because per-simulation host-device round trips cannot be amortized without within-episode leaf batching. Virtual-loss leaf batching does push the host pipeline to 163 ep/s (1.07× the kernel), but only by degrading the search itself: at leaf batch 32 the win rate vs. GD falls from 0.757 to 0.531 and the visit-distribution TV rises to 0.62; softening virtual loss to 0.25 recovers only 0.712. Per-component profiling shows the GD rollout evaluation dominates (∼75% of the host pipeline's evaluator wait time; policy priors and WP scoring share the rest), and that under the submission's search the kernel's per-node cache of opponent GD distributions was never used, because leaves expanded only at our-team steps (measured cache-miss count 0); the revision's search uses it at every opponent node. This comparison covers only the submission's search, the only one the host pipeline implements, and was measured on the original GPUs with their host CPUs. It was not repeated for the revision's search: the host pipeline runs 256 CPU threads, and on the RTX 5090 machine's 16 cores it reaches 9 episodes/second, so its speed there says more about that machine's CPU than about the kernel. The remaining advantages of the fused kernel are operational: 0.12 s startup, a deterministic ∼270 MB footprint, and no host thread pool or Python runtime at inference.

II. VALIDITY OF THE INTERACTION METRICS

The main paper's interaction-awareness metrics are built from the pairwise synergy and counter deltas (Section III of the main paper), which subtract from each observed pair win rate an independence expectation formed from the two heroes' individual win rates. We validate that these deltas carry real outcome signal by scoring each side of real drafts with the exact metric implementations against the game's actual outcome. Pooled naively, the decile curves are not monotone: subtracting the individual-strength expectation makes the deltas strongly anti-correlated with team hero strength (team synergy delta against team mean hero win rate), so teams of individually strong heroes receive negative deltas while still winning more. We therefore control for that component, computing deciles within tier and own/opposing team-strength strata.

The submission ran this check on the test replays with the external statistics, which contain those replays' own outcomes (first row of Table II). On 260,539 post-snapshot games, which were uploaded after the statistics were pulled and so cannot be in them, the same statistics give spreads of +8.8pp (synergy) and +9.2pp (counter), both monotone across all nine steps. The self-including spreads are 2.5pp and 1.8pp larger, about 30% and 20% of the clean values, though the two figures come from different game sets. Statistics built only from our training split give the same qualitative result on both game sets. The interaction deltas measure an outcome-relevant property of a composition beyond individual hero strength, which is the component they are designed to isolate.

III. STATISTICS BEHIND THE INTERACTION METRICS

The interaction metrics read pairwise win-rate statistics, and the main paper reports them with the community aggregates. Does the ordering of agents depend on that choice? We score the same benchmark drafts of every MCTS configuration (Table V of the main paper; root temperature 1) with two statistics sets: our own training-split statistics, which the leak-free value functions also read, and the external aggregates. Same policies, same drafts; only the lookup tables differ.

The counter ordering of the eight configurations is the same under both sets (Spearman 0.98), with a level shift of about 0.1. The synergy ordering is not (Spearman 0.55). Most of the disagreement is the featureless agent K, which is the least synergistic configuration under our statistics and mid-table under the external ones, and the in-sample agents, whose value function read our statistics and which score higher under them. Synergy levels therefore carry information about which statistics an agent was built on; we compare synergy across agents only under one statistics set, and the main paper uses the external aggregates for every row.

IV. External Validation on Coordinated-League Drafts

The main paper's behavior policy is uncoordinated ranked-ladder play, and its Limitations note that every evaluation shares that corpus. As an external test, we evaluate the frozen snapshot-trained win-probability models on drafts from the Nexus Gaming Series (NGS), an amateur organized league in which persistent five-player teams draft together with dedicated captains. Scanning the Heroes Profile NGS archive yielded 11,132 games, of which **11,034** pass filtering (patch 2.55, complete 16-step draft, known winner), spanning 2021–2026 across ten season clusters, the same patch era as our ladder corpus.

a) Coordinated drafts are more meta-concentrated.: A natural hypothesis is that coordinated teams, freed from solo-queue comfort picking, explore more of the hero pool. The data show the opposite, at both the hero and the composition level. Hero level: NGS pick entropy is 5.92 bits vs. the ladder's 6.25 (effective pool 60.5 vs. 75.9 heroes), the top-10 heroes take 33.5% of picks vs. 25.1%, and the off-meta share (picks outside the ladder top-30) is 39.8% vs. 43.3%; hero pick-rate rank correlation between the populations is 0.74. Composition level: NGS fields 78 distinct role compositions vs. 238 on ladder, with composition entropy 2.69 vs. 3.06 bits, the top-5 compositions covering 84.3% vs. 78.6% of teams, and 90% coverage reached in 9 vs. 12 compositions. The modal composition is the same standard core in both populations (bruiser/healer/double-ranged/tank: 53% vs. 50%); NGS's second-most-common composition is a double-tank variant (12.3%). Our reading: coordinated teams converge on a known-good playbook, while ladder "diversity" partly reflects uncoordinated comfort picking and error. This fits the main paper's description of the behavior policy as shaped largely by social habit. Coordinated teams draft more narrowly, so the ladder's wider spread reflects comfort picking more than exploration.

b) Win-probability transfer.: Scored at the game level with symmetrized predictions, the leak-free enriched model reaches 55.4% accuracy on NGS (95% CI $\pm$0.9pp) against 57.5% on the held-out ladder test set; the leak-free naive model also scores 55.4%. NGS games are in neither our corpus nor the Storm League aggregates, so no model has self-inclusion here. The ladder-to-NGS drop is 2.1–2.2pp for both models. The submission compared NGS with a leaky ladder figure (58.0% on our test set) and reported a larger drop for the enriched model; its enriched model scores 55.5% on NGS, with calibration slope 0.75 against 0.85 for the leak-free enriched model and 0.91 for the leak-free naive model. The submission also reported that the degradation concentrates in off-meta drafts. Splitting NGS drafts into quartiles by off-meta pick share, the leak-free models show no such trend: the enriched model scores 56.8%, 54.3%, 55.7%, and 54.7% from most- to least-meta quartile (about 2,760 games each, CI $\pm$1.9pp), and the naive model 57.0%, 53.9%, 55.9%, and 54.8%. We therefore do not claim that the degradation concentrates in off-meta drafts. 99.98% of NGS team compositions fall within the composition sets observed in the ladder-era aggregate statistics, so this shift re-weights familiar structures and does not reach the never-observed compositions. One caveat caps interpretation: NGS outcomes also reflect persistent team-strength imbalances that the models never observe (the same organized teams meet repeatedly), which lowers attainable accuracy in a way not directly comparable to solo-queue ladder play.

V. Draft Concentration and Out-of-Distribution Extrapolation

How far do the agents drift from human play? We measure the operating-point agent (F, 5 seeds $\times$ 1,000 benchmark drafts, mid tier, 200 inference simulations, root temperature 1) and, for comparison, J (800 training simulations, 5 seeds).

a) Concentration: Each F seed uses 22–26 distinct heroes at 3.24–3.52 bits of pick entropy; pooling the five seeds gives 33 heroes and 3.54 bits, so the seeds converge on one head. Its seven favorite heroes are Samuro, Rehgar, Hogger, Deathwing, Falstad, Artanis, and Abathur; J's head overlaps heavily (Hogger, Samuro, Rehgar, Deathwing, Illidan, Cho, Auriel) and is slightly more diverse (26–31 heroes per seed, 3.61 bits pooled).

b) Are the favorites' interactions real?: For all 21 pairs and 35 trios of the seven favorites we compare the win rate of real teams containing them with the independence expectation from the heroes' own win rates (the paper's synergy convention). In our training split the favorite pairs beat independence by +0.36pp (n = 273,540 team-games, z = 3.8) and the trios by +1.16pp (n = 17,771, z = 3.1). On 414,926 post-snapshot games (including the drifted 2.55.17 builds), which no model saw, the pairs beat it by +0.09pp (n = 60,675, z = 0.4) and the trios by +0.75pp (n = 4,013, z = 0.9). Neither edge replicates. J's favorites do somewhat better forward (+0.55pp pairs, z = 1.9; +1.90pp trios, z = 1.1), still short of significance. The submission's statement that search "exploits precisely the interactions the data certifies", based on in-sample checks only, is withdrawn.

c) Where the agent leaves the data: None of F's ten most frequent five-hero teams (together 41.2% of its drafts) occurs even once in the 1.94M-game corpus. One level up it stays in familiar territory: 96.9% of its drafts land on role compositions admitted to our own composition table (at least 50 games in the tier label) and 47.4% on compositions with at least 1,000 games.

VI. Ensemble Epistemic Uncertainty of the Value Model

Do functionally different value models agree on the drafts the tournament judged? We train a 20-member leak-free ensemble on our out-of-fold features (validation early stopping): 8 full enriched models varying architecture (256$\rightarrow$128 or 512$\rightarrow$256$\rightarrow$128), training-data half, and seed; 9 models each dropping one enriched feature group; and 3 identity-only models. For each draft we compute the variance of the 20 symmetrized win probabilities.

The operating-point agent's benchmark drafts draw near-human disagreement (1.24$\times$ the human median, 10.6% beyond

TABLE IV
PER-DRAFT VARIANCE OF THE 20-MEMBER LEAK-FREE ENSEMBLE. RATIO: MEDIAN RELATIVE TO HELD-OUT HUMAN DRAFTS; >P95: SHARE OF DRAFTS ABOVE THE HUMAN 95TH PERCENTILE.

Draft set	Med. $\times 10^3$	Ratio	>p95
Held-out human (n=2000)	0.42	1.00	(5.0%)
F agent (n=2000)	0.53	1.24	10.6%
Synthetic degenerate (n=500)	1.58	3.74	64.8%
Degenerate probes (n=6)	2.21	5.22	5/6
Tournament drafts per strategy (n=2000 each)			
MCQ	1.50	3.55	65.2%
Gourdeau estimator	0.95	2.24	39.0%
Enriched greedy	0.87	2.06	34.0%
Constrained greedy	0.86	2.03	34.4%
MCTS, no features	0.78	1.85	29.8%
MCTS	0.75	1.78	26.1%
Constrained MCTS	0.72	1.71	24.4%
CQL enriched	0.61	1.45	24.7%
CQL naive	0.60	1.42	21.4%
GD	0.59	1.40	21.6%
G&A discriminator	0.55	1.31	22.2%

TABLE V
HELD-OUT QUALITY ON HALF B OF VALUE FUNCTIONS TRAINED ON HALF A. "OWN-SPLIT VAL.": WHAT A LEAKY PIPELINE REPORTS ABOUT ITSELF (LAST EPOCH).

Proxy statistics	Games	Own-split val.	Held-out	Slope
in-sample, 1/8 of A	119K	67.8%	53.3%	0.17
in-sample, all of A	952K	59.7%	56.2%	0.62
out-of-fold, all of A	952K	56.2%	57.0%	1.02
hero identity only	952K	56.5%	57.1%	1.04

the human 95th percentile). Degenerate drafts are sharply elevated (3.7× for synthetic unseen compositions, 5.2× for the fixed probes). Every strategy's tournament drafts but MCQ's sit between 1.31× and 2.24×, below the degenerate regime; MCQ, all of whose teams are degenerate, sits in it (3.55×). The anchored strategies are priced most confidently and the strategies that search a value function off the human manifold highest. The tournament's verdicts were therefore issued where the model family mostly agrees, with the least agreement among searchers on the Gourdeau estimator's drafts (39.0% beyond the human 95th percentile).

VII. LEAKAGE MECHANISM: A WITHIN-SNAPSHOT SPLIT EXPERIMENT

To isolate how in-sample statistics affect a value function and the search that optimizes it, without any balance drift, we split the ranked snapshot by a salted replay-level hash into halves A and B. Proxies, their statistics, their early stopping, the behavioral-cloning prior and opponent of the search, and any search against them use only half A; references are built only from half B (three out-of-fold enriched judges, a hero-identity judge, and a cross-fitted realized-outcome index), plus the post-snapshot index and the Quick Match judge.

Table V shows the mechanism. In-sample statistics make the value function overconfident on games outside them (slope 0.62 at 952K games, 0.17 at 119K) and less accurate, while

TABLE VI
HELD-OUT ACCURACY AND CALIBRATION SLOPE OF EACH REFERENCE ON GAMES NONE OF THEM TRAINED ON: THE PAPER'S 38,869 TEST REPLAYS, AND 154,387 GAMES ON THE DRIFTED 2.55.17 BUILDS (R17 IS FIT ON THOSE GAMES AND OMITTED THERE).

	Test (snapshot)		2.55.17	
Reference	Acc.	Slope	Acc.	Slope
gN (3 OOF enriched judges)	55.4	0.92	55.5	0.88
gN-naive	55.7	0.95	55.8	0.90
RN (realized index)	54.7	1.04	54.8	0.98
R17 (realized index, drifted)	53.7	0.71	–	–
QM2026	54.0	0.55	53.6	0.53
QM2021	54.7	0.40	52.6	0.25
Leak-free naive, 1 seed (in corpus)	57.8	1.23	56.3	0.96

reporting inflated accuracy on its own validation rows. Out-of-fold statistics remove the effect. Width, depth, dropout, weight decay, and training length each move held-out accuracy by at most 0.1pp, except training without regularization for 10 or 40 epochs (−0.3pp and −1.0pp; slope 0.57 and 0.40). Leakage dominates every capacity knob.

Search then exploits the error. With a behavioral-cloning prior and argmax root selection, raising search from 1,024 to 16,384 simulations lifts the in-sample proxy's own score from 0.754 to 0.834 and its half-B judges from 0.644 to 0.671; the out-of-fold proxy, whose own score (0.680 → 0.740) tracks the judges almost one for one, reaches 0.660 → 0.721 on the same judges and higher values on every realized index (half-B index 0.659 vs. 0.628, post-snapshot index 0.619 vs. 0.592). A small in-sample proxy (1/8 of A) shows classic over-optimization: its own score climbs to 0.941 while the half-B judges stop improving after 4,096 simulations (0.611, then 0.610). A leak-free proxy trained on 1/8 of the data beats an in-sample proxy with 8× the data on both realized indices (0.668 vs. 0.628 and 0.637 vs. 0.592 at 16,384 simulations).

The paper's own self-play pipeline on half A (2 seeds per cell, 300K episodes each, the behavioral-cloning bootstrap and the GD opponent trained on half A) shows the same split. Raising training simulations from 200 to 800 moves the in-sample agents' half-B judges from 0.686 to 0.607 and their degenerate rate from 27% to 75%, and lowers both realized indices by 0.03; the out-of-fold agents gain 0.016 under the same judges at a 3–4% degenerate rate.

Two diagnostics that did not detect the problem: the per-draft disagreement of a 4-seed ensemble trained on the same statistics stays between 0.005 and 0.007 at every search pressure (seeds share the leak), and pessimistic lower-confidence-bound search using that disagreement lifts the in-sample ensemble's half-B judges by at most 0.012 (0.639 → 0.651), well short of the out-of-fold ensemble (0.661). Held-out calibration slope did detect it, which is why the main paper reports it for every value function and judge.

VIII. CALIBRATION OF THE INDEPENDENT REFERENCES

The post-snapshot judges and the realized index are close to calibrated on snapshot games. The Quick Match judges are strongly overconfident on ranked drafts (slopes 0.40–0.55):

TABLE VII
REALIZED MINUS PREDICTED WIN RATE (PP) OF REAL TEAMS BY TYPE, ON 1.94M SNAPSHOT GAMES NO REFERENCE TRAINED ON (SE 0.2, 0.3, 1.0, 0.2PP BY COLUMN). TYPES COVER 1.37%, 0.57%, AND 0.06% OF TEAMS. V2: AFTER THE STRUCTURE CORRECTION. ΔLL: V2 MINUS V1 LOG LOSS ($\times 10^{-4}$).

Reference	No heal.	No front.	Stack	Any	Any, v2	ΔLL
gN	−2.6	−1.4	−1.2	−2.2	+0.6	−0.4
gN-naive	−5.9	−3.8	−9.8	−5.5	−0.2	−2.8
RN	−5.0	−4.1	−10.8	−4.9	+0.4	−2.3
R17	−6.5	−6.1	−11.9	−6.5	−0.2	−3.9
QM2026	−5.9	−3.1	−9.4	−5.2	+0.1	−2.7
QM2021	−6.8	−3.4	−7.5	−5.8	+0.5	−3.3
Consensus	−4.8	−3.1	−7.8	−4.4	+0.2	−1.9

their probabilities are too extreme by a factor of about 2, so the main paper uses them for ordering only. All references are well below the 57.8% of an in-corpus model (the selected seed of the leak-free naive model; the main paper's 57.6% is its three-seed mean), the price of independence: smaller training sets, a different time window, or a different game mode.

IX. COMPOSITION BLIND SPOT AND CORRECTED REFERENCES

a) Audit: A drafter that picks the available hero with the highest win rate scored 0.677 under the production seed-selection judge (a realized index on 90 days of games), above both trained policies (0.644, 0.650), although 55% of its teams were degenerate. Table VII measures every reference on the 1,943,338 snapshot games, which none of them trained on, as realized minus predicted win rate per team type (no healer, no frontline, three or more of a stacking role; exclusive, in that order). Normal teams are calibrated to within 0.1pp. Degenerate teams lose 5–7pp more often than RN, gN-naive, R17, and both Quick Match judges predict; the gap survives a global slope and intercept recalibration of each judge. gN, which reads role-count features, misses 2.2pp. The post-snapshot no-drift games give the same picture with held-out predictions (gN from two-fold refits of its recipe: −2.8pp; RN cross-fitted: −5.2pp; consensus −4.6pp). RN is blind because its composition term is a per-role-multiset win rate shrunk toward 50% with a 200-game prior, and degenerate multisets hold a median of 3 (high tier) to 9 (low tier) games; gN-naive must learn the penalty from hero identities alone; the QM judges learned from a mode with a different team-assembly process.

b) Selection or cause: Weaker players might field broken teams. We joined 897,779 games (snapshot games from June 2025 and post-snapshot games with all ten player records) to causal per-player, per-hero skill estimates from the previous day (a Gaussian-process model over each player's heroes; skill is estimated with degenerate-team games left out, so it cannot absorb the penalty itself) and to pick order. Degenerate teams are weaker (summed skill −0.3pp vs. +1.0pp for normal teams), but matchmaking pairs them with weaker opponents too. Against RN, the penalty is −4.79pp with no controls, −4.62pp with the team skill difference, −4.83pp adding skill by pick position and first-pick side, and −4.65pp adding party and never-played-hero counts (bootstrap SE 0.32pp; SE of the change 0.04pp). At most about 0.2pp of the 4.8pp is selection on these measures. Unmeasured differences between players remain possible.

c) Correction: For each reference J we fit $p^* = \sigma(\mathrm{logit}\, p_J + \beta_J^\top (s_{\text{own}} - s_{\text{opp}}))$ with s the three structure indicators (non-exclusive) and p_J fixed, on the 260,539 post-snapshot no-drift games, using predictions J makes for games it did not train on (above). The fitted no-healer, no-frontline, and stack coefficients (logits) are −0.10, −0.13, −0.14 (gN); −0.21, −0.20, −0.38 (gN-naive); −0.18, −0.25, −0.38 (RN); −0.21, −0.19, −0.36 (QM2026). The correction is antisymmetric in the two teams, so symmetrized scores stay symmetrized. A version with the skill controls gives coefficients within 0.14 and over-corrects the snapshot games by 0.8pp, consistent with little selection; we use the plain fit. On games after the 2.55.17 balance patch the penalty is smaller and the corrected references over-charge degenerate teams by about 2pp. The terms are learned from human degenerate teams, 93% of which have exactly one defect (68% no healer); the MCTS agents' degenerate teams have exactly one defect in 93–100% of cases and are 80–98% no-healer teams, the best-covered type. A structure nobody fields, such as five tanks, receives only the sum of its indicators.

d) Effect on the results: Under the corrected references, the benchmark contrasts behind sim scaling, the feature gap, training length, and the operating point change by at most 0.007 (Table VIII): GD opponents sampled at temperature 1 field degenerate teams too, so the correction charges both sides of the MCTS benchmark. The leak-control contrasts sharpen by up to 0.011 because the leaky 800-simulation agent fields 27% degenerate teams. In the tournament the constrained agents gain (constrained vs. unconstrained MCTS 0.497 → 0.513, masked vs. unmasked greedy 0.500 → 0.511, ±0.004 and ±0.003), the featureless MCTS agent falls from first to second (0.602 → 0.607, behind constrained MCTS at 0.597 → 0.612), the Gourdeau estimator (45% degenerate) drops from 0.534 to 0.521, and MCQ from 0.370 to 0.295. Rank agreement of the Quick Match judges within the top six falls (0.89 → 0.66), because the correction moves constrained MCTS ahead of K and the Gourdeau estimator behind both greedy agents, orderings those judges do not share.

e) Composition table of gN: gN reads a role-composition table for 4 of its 283 inputs. Any table built from the community aggregates overlaps our snapshot, so gN builds its own from its post-snapshot games (out of fold for its training rows) and shares no statistic with any agent. It reaches 55.4% test accuracy with slope 0.92.

X. JUDGE ATTENUATION

An independent judge is a noisier model than the value function an agent optimized, so a real gain shrinks under it. As a benchmark for that shrinkage we regress each judge's prediction on the value function's prediction over the 38,869 test replays, which neither trained on (Table IX). With no over-optimization, a gain d under the value function should appear as about slope $\times d$ under the judge, if the value function's

TABLE VIII
PAPER CLAIMS UNDER THE UNCORRECTED (V1) AND CORRECTED (V2) REFERENCES. MCTS BENCHMARK CONTRASTS, ROOT TEMPERATURE 1 (5 SEEDS; 3 FOR E AND THE IN-SAMPLE ROWS); ±: SEED SE OF V2.

	gN		RN	
Contrast	v1	v2	v1	v2
F − B	−.005	−.003	−.005	−.001 ±.006
J − F	+.003	+.003	+.001	+.001 ±.006
F − B, in-sample	−.004	−.004	−.002	−.002 ±.005
J − F, in-sample	−.012	−.016	−.006	−.013 ±.006
J in-sample − J	−.023	−.029	−.012	−.023 ±.006
F − K	−.001	+.002	−.008	−.002 ±.006
E − B	−.007	−.006	−.009	−.007 ±.004
F − E	+.001	+.003	+.003	+.007 ±.004
F, argmax − T=1	+.002	+.002	+.002	+.002 ±.001

TABLE IX
SLOPE OF EACH COMPOSITION-CORRECTED JUDGE ON THE VALUE FUNCTION'S PREDICTION, TEST REPLAYS (BOOTSTRAP SE ≤ 0.01).

Value function	gN	gN-nv.	RN	QM26	Cons.
Leak-free enriched	0.59	0.66	0.45	0.49	0.55
Submitted enriched	0.53	0.60	0.40	0.44	0.49

errors on agent drafts resemble its errors on human drafts. The submitted value function's statistics contain the test games, so its slopes are a rough guide.

Against this benchmark, neither doubling of training simulations transmits in the leak-free pipeline (200 → 400: own +0.017, predicted gN +0.010, observed -0.003 ± 0.004; 400 → 800: own +0.011, predicted +0.006, observed $+0.003 \pm 0.004$). With in-sample statistics the own score rises faster (+0.043, then +0.026) while gN falls (-0.004 ± 0.004, then -0.016 ± 0.005): over-optimization.

XI. FEATURE-GROUP SWEEP (FRACTIONAL FACTORIAL)

The submission's sweep trained all 2^{10} subsets of ten feature groups at two architectures and two learning rates (4,096 configurations) with features that contained the test games' outcomes. The revision re-runs the question with leak-free statistics using a 2^{10-3} fractional factorial: base factors map_type, role_counts, hero_wr, team_avg_wr, hero_map_wr, map_delta, pairwise_counters (A–G), and generated factors pairwise_synergies = ABCD, counter_detail = ABEF, synergy_detail = ACEG. Every word of the defining relation has length at least 5 (resolution V), so main effects are unaliased with each other and with all two-factor interactions, and each main effect equals the full-factorial marginal the submission reported, up to four-factor and higher interactions. Each of the 128 configurations trains one 256→128 model (seed 42, learning rate 5×10^{-4}, validation early stopping). Effect SE, from the residual after fitting main effects: 0.02pp (test accuracy), 0.01pp (post-snapshot accuracy), 4×10^{-5} (post-snapshot log-loss).

The four pairwise groups lose 0.12–0.22pp each relative to the submission, which is where self-inclusion was concentrated (the submission also averaged over two architectures and two learning rates, a smaller difference). Role counts, which read no statistics, keep their effect. Every effect is below 0.15pp in absolute value on both held-out sets.

TABLE X
MAIN EFFECT OF EACH FEATURE GROUP (WITH MINUS WITHOUT), LEAK-FREE STATISTICS. SUBMISSION: THE REPORTED FULL-FACTORIAL MARGINAL WITH LEAKY STATISTICS.

Group	Test	Post	Post LL$\times 10^3$	Submitted
role_counts	+0.11	+0.13	−0.72	+0.16
pairwise_counters	+0.04	+0.03	+0.08	+0.20
pairwise_synergies	+0.03	+0.02	+0.01	+0.15
map_type	+0.02	+0.01	−0.02	+0.01
map_delta	−0.00	+0.02	+0.04	+0.09
synergy_detail	−0.00	−0.05	+0.36	+0.16
team_avg_wr	−0.03	−0.04	+0.14	−0.05
counter_detail	−0.07	−0.08	+0.69	+0.14
hero_wr	−0.08	−0.11	+0.41	−0.16
hero_map_wr	−0.09	−0.10	+0.47	−0.02

XII. OFFLINE RL: SWEEPS AND ROBUSTNESS

The main paper's Table IV shows one or two cells per offline-RL family. The full sweeps, trained on our split with naive (identity) features unless noted and selected on validation games, all end at an anchored policy. CQL across $\alpha \in \{0.1, 0.5, 1, 2, 5\}$ (5 seeds × 1000 drafts each): 97.5–99.7% healer and 1.9–9.3% degenerate, best at $\alpha = 2$ (99.6%, 1.9%). BC-regularized CQL across $\beta \in \{0.1, 0.5, 1, 2\}$: near-perfect safety (99.5–100% healer), high GD agreement (70–74%), negative synergy. MCQ across $\tau \in \{0.4, \ldots, 0.8\}$: for $\tau \geq 0.6$ the penalty is inactive and the policy uses the same 8 heroes regardless of context; $\tau = 0.5$ gives the dead-network policy of the main text (all 128 units of hidden layer 3 and 255 of 256 of layer 2 output zero on 200,000 held-out states; at $\tau = 0.6$, 126 of 128 units of layer 3 are dead; a CQL $\alpha = 1$ control has 26 and 13 dead units). IQL across a $\tau \times \beta$ grid of six cells: statistically identical cells, Q-network held-out loss 0.6948 against $\ln 2 = 0.6931$ for a constant.

Capacity and target-update robustness. At $\alpha = 1$ we crossed three network sizes (256→128→64, 512→256→128, 1024→512→256) with five target-network update rates ($\tau_{\text{target}} \in \{0.001, 0.005, 0.01, 0.05, 0.1\}$; 200 drafts each). Across all 15 configurations the collapse signals are stable: healer 98.5–100%, degenerate 1.0–5.0%, 46–57 distinct heroes, entropy 4.44–4.59 bits, GD agreement 68.4–73.9%, synergy −0.13 to +0.03. No configuration is interaction-aware, and the update rate is inert, as expected with Monte Carlo returns (no bootstrapping).

CQL with enriched features. Trained on out-of-fold enriched transition features and evaluated with deploy statistics, enriched CQL behaves like naive CQL: 3.4–3.6% degenerate, synergy −0.18, entropy 4.65–4.67 bits, GD agreement 68.4–70.4%. Under the independent references these agents score 0.49–0.52 against behavioral-cloning opponents.

XIII. EXPERT STUDY: DESIGN AND ANALYSIS NOTES

a) Registration and materials: The plan was registered before data collection (https://osf.io/fspuc). One change was

made before any invitation: the rating page stopped advancing automatically after the last answer, because in a test run a click meant for one item could submit the next; raters press Next after all three answers. The item pool, analysis code, frozen ratings export, and outputs are in the repository (`training/paper1_revision/expert_analysis`, `results/expert_v6/analysis_final`).

b) Raters and gates: Fourteen slots, each a fixed sequence of 240 items: 8 screening items (a real winning draft against a role-broken team) and 40 calibration games shared by all raters in positions 1–48, then 60 machine pairs and 129 real games, with 3 attention checks at positions 121, 181, and 231. One slot was re-issued to a new rater with the same assignment before that slot's first rating. Eleven raters completed 240 items. Gates: at least 7 of 8 screening items, at least 2 of 3 attention checks, and a median response time of at least 10 seconds. Two completers failed the screener (6 of 8 each); nine passed every gate. Three raters stopped at 7, 24, and 63 items. Gates are applied as written, so a rater who never reached a gate's items cannot pass it and all three are excluded; holding them only to the items they reached admits the rater with 63 items; no conclusion changes, and every rate moves by 0.01 or less.

c) Choices the plan did not settle: (1) The rule for incomplete raters above. (2) "Complete" in the condition for the low-rank-experience contrast (at least three such raters complete) means 240 ratings, regardless of gates; three completed and all three passed. (3) A model fit counts as failed, triggering the registered fallback, only if it errors or the fitting software reports non-convergence; a singular fit is kept. (4) The real-game accuracy test had no registered fallback; the low-tier fit gave a convergence warning, and we used the first fallback of the primary test (dropping the rater random effect). (5) In the majority-vote fallback, items whose ratings split evenly are dropped (no fallback was needed). (6) The permutation fallback of the robustness test uses 100,000 permutations (not needed). (7) The tournament ordering for the ranking test is each strategy's mean composition-corrected label win probability over all its tournament drafts. (8) The unfamiliarity covariate is standardized over the analysis items of each subset. (9) The patch-2.57 exclusions are applied at the primary 0.02 near-tie threshold.

d) All registered tests: Mixed-effects logistic models with crossed rater and item random effects (Laplace approximation, Wald intervals); the slider model is linear with a random slope per rater. Agreement with the model label on clear pairs: 0.50 (0.43–0.57); at near-tie thresholds 0.01 and 0.05, 0.52 and 0.49; with each single value function as the label, 0.50, 0.50, and 0.50; with the uncorrected label, 0.50. Anchored-vs-searching pairs: 0.33 (0.21–0.48). Robustness to unfamiliarity: slope $+0.24$ per SD (CI -0.02 to 0.50) for the maximum, $+0.32$ ($+0.06$ to 0.58) for the mean, and $+0.17$ (-0.08 to 0.42) for the matchup-level ensemble variance; all three meet the registered non-inferiority margin (-0.25 per SD). Real-game accuracy by tier: 0.60, 0.49, and 0.53 (Holm-adjusted p = 0.004, 0.85, 0.85). Slider calibration slope 0.00 (-0.22 to 0.22). Low-rank-experience contrast: -0.09 log-odds (-0.55 to 0.36), one-sided $p = 0.66$. Ranking: Spearman $\rho = -0.19$ between the experts' Bradley–Terry strengths and the tournament order (exact one-sided $p = 0.71$ over all 10! orderings).

e) Model accuracy on the same real games (descriptive, post hoc): Not in the registered plan. On the 570 anchor games, the model label (the mean of the three corrected value functions) picks the winner in 60.0% (52.9–66.7), 54.2% (47.1–61.1), and 55.8% (48.7–62.7) of games at low, mid, and high tier, 56.7% (52.6–60.7) overall; the enriched value function alone scores 64.7%, 55.3%, 55.8%, and 58.6% overall (Wilson 95% intervals in parentheses, 190 games per tier). Pooled over raters, experts picked the winner in 58.7%, 49.4%, and 52.2% of judgments, 53.4% overall. With 190 games per tier the comparison is loose; above the lowest tier the experts are at chance and the models near the 57–58% that draft-only models reach on held-out games.

f) Patch 2.57: Patch 2.57 went live on 2026-09-28, after every real game in the pool and before the machine drafts were generated by agents trained on earlier games. Every item and the instructions state the patch (2.55.17). Dropping items with any of the seven heroes whose balance changed in 2.57 leaves agreement at 0.49 (195 clear pairs), anchored-vs-searching agreement at 0.33 (97 items), and pooled real-game accuracy at 0.60, 0.45, and 0.51 by tier. A broader cut that also drops four heroes with gameplay bug fixes and two changed maps leaves 57 clear pairs (agreement 0.55) and too few anchored-vs-searching pairs (16) to estimate.